# Day 11 — Excel Tables & Dynamic Formulas

*Learning notes • Examples • Formula reference*

## Learning Objectives

- Create and manage Excel Tables.
- Use structured references instead of ordinary cell addresses.
- Understand dynamic ranges for growing datasets.
- Create and use Named Ranges.
- Use UNIQUE, SORT, and FILTER in compatible Excel versions.

## 1. Excel Tables — Deeper

An Excel Table is a structured range of data with headers, filtering, consistent formatting, and formulas that can adapt as records are added.

### Create a table

1. Select the data, including the header row.
2. Press Ctrl + T.
3. Confirm "My table has headers," then click OK.
4. On Table Design, give the table a clear name, such as SalesData.

Example table (named SalesData):

| Salesperson | Country | Sales |
|---|---|---|
| **Rahul** | Kenya | 25,000 |
| **Priya** | Nigeria | 32,000 |
| **Amit** | India | 28,000 |

Tables can expand when records are added to them, and calculated-column formulas can fill down automatically.

## 2. Structured References

Structured references use table and column names instead of cell addresses. They are readable and can adapt when the table grows.

| Purpose | Formula | Meaning |
|---|---|---|
| **Total the Sales column** | =SUM(SalesData[Sales]) | Adds the values in the table's Sales column. |
| **Whole Sales data column** | SalesData[Sales] | Refers to the Sales data column. |
| **Current row's Sales** | [@Sales] | Inside the table, refers to Sales in the current row. |
| **10% commission** | =[@Sales]*10% | Calculates 10% of each row's sales. |

Example: =SUM(SalesData[Sales]) is generally easier to maintain than =SUM(C2:C4), because a table reference can include additional rows added to the table.

## 3. Dynamic Ranges

A dynamic range adjusts as data changes. A fixed range such as B2:B4 will not include a new record in B5 unless the range or formula is expanded.

Fixed range: =SUM(B2:B4)

Table-based formula: =SUM(SalesData[Sales])

Practical tip: for most growing business datasets, use an Excel Table instead of manually extending ranges. Make sure new records are added to the table.

## 4. Named Ranges

A Named Range gives a cell or range a meaningful name, making formulas easier to understand.

Example: if B2 contains a sales target, name B2 SalesTarget.

Instead of =A2/B2, use =A2/SalesTarget.

If A2 contains a sale amount and a named cell TaxRate contains 18%, use =A2*TaxRate to calculate tax.

### Create a Named Range

5. Select the cell or range.
6. Click the Name Box to the left of the formula bar.
7. Type a name such as SalesTarget and press Enter.

Alternative: select the range and choose Formulas → Define Name.

### Naming rules

- Avoid spaces; use Sales_Target or SalesTarget.
- A name must not look like a cell reference such as B2.
- Choose descriptive names such as MonthlyTarget, TaxRate, or CommissionRate.
- Use Formulas → Name Manager to review or edit names.

## 5. Dynamic Formulas and Dynamic Arrays

Dynamic array formulas can return multiple results that spill into neighboring cells. These functions are available in Microsoft 365 and supported newer versions such as Excel 2021 or later; they may not work in older versions.

| Function | Example | Purpose |
|---|---|---|
| **UNIQUE** | =UNIQUE(SalesData[Country]) | Returns distinct country names. |
| **SORT** | =SORT(UNIQUE(SalesData[Country])) | Sorts the distinct country names. |
| **FILTER** | =FILTER(SalesData,SalesData[Country]="Kenya","No records") | Returns rows where Country is Kenya; shows No records if there are no matches. |

### Important notes

- Enter a dynamic array formula in one cell; Excel spills the results into adjacent cells.
- Keep the spill area empty or Excel may show #SPILL!.
- Unsupported functions may show #NAME? or be unavailable.
- Tables help source ranges adapt; dynamic array functions return changing sets of results.

## Quick Formula Reference

| Goal | Formula |
|---|---|
| **Total sales** | =SUM(SalesData[Sales]) |
| **10% commission for current row** | =[@Sales]*10% |
| **Tax using a named rate** | =A2*TaxRate |
| **List unique countries** | =UNIQUE(SalesData[Country]) |
| **Sort unique countries** | =SORT(UNIQUE(SalesData[Country])) |
| **Filter Kenya records** | =FILTER(SalesData,SalesData[Country]="Kenya","No records") |

## Practice Questions

8. Your table is named Orders and has an Amount column. Which formula totals the Amount column?
9. What does [@Sales] mean inside a table?
10. Why is =SUM(SalesData[Sales]) useful when records are added?
11. If B2 is named TaxRate, which formula calculates tax on A2?
12. Which function returns distinct country names?
13. Which function returns only rows that match a condition?

### Answer Key

- =SUM(Orders[Amount])
- The Sales value in the current row.
- The table reference can include new rows added to the table.
- =A2*TaxRate
- UNIQUE
- FILTER

## Summary

Use Excel Tables for growing datasets, structured references for readable formulas, Named Ranges for meaningful references, and dynamic array functions for automatically updating lists and filtered results. Confirm your Excel version supports the functions you plan to use.